

ISSS602 Data Analytics Lab

Data Analytics Project

(AY2025-26T2.1)

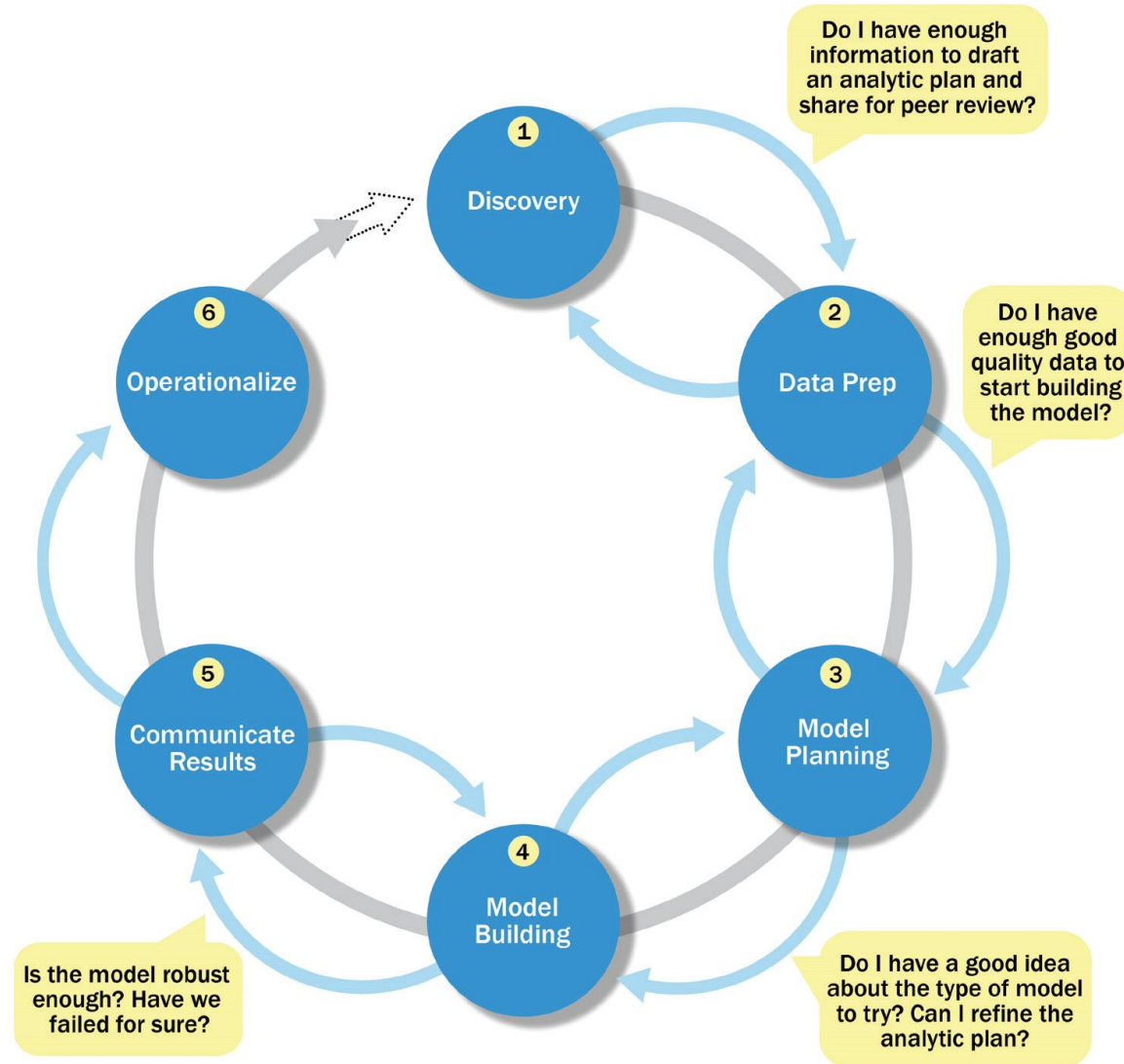
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Aims

To provide students the opportunity:

- to conduct an analytics project by using real world data sets.
- to explore new data analytics approach(es) or/and technique(s) beyond the lessons.
- To gain indepth hands-on experience on the data analytics software tool(s) use in this course.

Data Analytics Lifecycle



Learning Outcomes

Upon completion of this project, student will be able to:

- Identify and define a data analytics project.
- Plan a data analytics project.
- Conduct a data analytics project by using appropriate data sets, data analytics techniques, and data analysis toolkit(s).
- Present the findings in the form of public poster session.
- Prepare a practice research paper highlighting the research contributions of the study.

Important deadlines

- Project proposal submission: 8th March 2026, by 23:59pm (Sunday, mid-night)
- Project Poster submission: 1st April 2026, by 23:50pm (Wednesday, mid-night)
- Poster presentation: 4th April 2026, 10:00am-11:30am. Venue: Alcove (SCIS1 basement 1)
- Final Submission: 5th April 2026 by 23:59pm (Sunday, mid-night)

Project topics?

- Will be released by 1st March 2026 23:59PM (mid-night)

Data Analytics Project Grades Summary

- Data Analytic Project 35%
 - Peer Evaluation (Minutes of Meeting) 15%
 - Project Poster 20%
 - Practice Research Paper 50%
 - Artifacts (data, metadata, etc) 15%

Poster specification

- Size = ISO A1 (594 × 841mm or 23.39 × 33.11inci)
- Resolution = 300dpi or above
- 7016 **pixels** x 9933 **pixels** (print resolution)
- File format = jpeg
- <https://support.microsoft.com/en-us/kb/827745>
- Name the poster as **ISSS602_AY2025-26T2_G0-Group00.**
- Submission: LMS Dropbox.

Poster grading criteria

- Clear communication of key aspects of solution
- Clear communication of analytical approaches
- Clear managerial communication of analysis results
- Quality of the poster

Data Analytics Poster



Predicting Status of Water Pumps in Tanzania

Choo Shao Ying

Jenifer Soo Shui Lian

Su Shiyan

Prof. Kam Tin Seong



Introduction

Water is an important basic need for all human beings, however in many developing countries, access to safe and clean water is not a commodity. In Tanzania where almost a third of the country is arid to semi-arid, clean, sanitary water is hard to come by if one does not live near one of the three major lakes that borders the country. Delivery of water to the people in Tanzania is a crucial aspect and thus the motivation behind determining the functionality of the water points in Tanzania.

Objective

In this project, water pumps in Tanzania and its characteristics would be analysed to predict if the pumps were functional, functional but needs repair or non functional. With an understanding of which water point can fail from the model, maintenance operations in Tanzania could possibly improve and clean, sanitary water made available to all communities across Tanzania.

Data Preparation

Recoding the data: Grouping levels of variables with low frequencies together

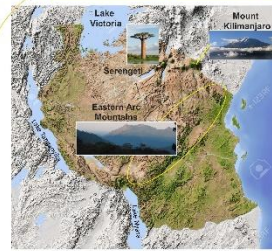
waterpoint_type_group	Frequency	waterpoint_recode	Frequency
cattle trough	116	communal standpipe	34625
communal standpipe	34625	hand pump	17489
dam	17489	improved spring	784
hand pump	17489	other	6393
improved spring	784		
other	6393		

Binning: Numeric variables with a lot of 0s were binned.

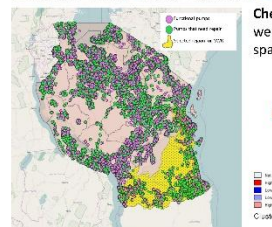


Missing Indicators are created for variables with missing data.

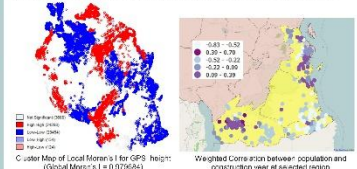
Methodology of Geographically Weighted Regression



The water pumps are distributed over the vast land of Tanzania, which spans a total area of 947,303 sq km. Due to this huge physical geography and topology differences across the country, there is a possibility that there may be local variations in the variables affecting the functionality of the water pumps' status. They may be masked using a global model that assumes spatial homogeneity. To address this problem, a binary class Geographically Weighted Regression (GWR) was used as an alternative prediction model; "functional needs repair" class will be combined with "non-functional".



Checking for spatial auto-correlation. Local Moran's I and weighted pair-wise correlations are used to check for spatial auto-correlation in the independent variables.



Weighting and Kernel: A fixed bandwidth using bi-square kernel is used for the weighting of the effect of the neighbouring data points at each location.

Interpolation at prediction locations. As GWR computes a set of coefficients which are unique to each location, there is a need to interpolate to find the coefficients at the prediction locations. This is done via a method called ordinary kriging.



Model Comparison

Multinomial logistic Regression and Decision Tree. Cut-offs for "non-functional" and "functional" class are set at 0.33 and 0.54 respectively. From the true positive results, the accuracy of rare cases "functional needs repair" has been boosted. Decision tree also gives better true positive rates compared to multinomial logistic regression.

Water Pump Status	Resampled Dataset		Non-resampled Dataset	
	Multinomial Logistic Regression	Decision Tree	Multinomial Logistic Regression	Decision Tree
TP - Functional	77.23%	73.35%	76.38%	82.56%
TP - Func Needs Repair	16.30%	22.18%	12.98%	13.48%
TP - Non Functional	70.42%	73.27%	71.52%	70.85%
Accuracy	70.17%	69.59%	69.9%	73.04%

Geography Weighted Regression. To find out if GWR could improve prediction results, a binary logistics regression is used as comparison. A threshold of minimum 80% true positive rate and 70% true negative rate is set as the balance point between unnecessary cost incurred to send maintenance personnel to a functional water pump vs the cost of people living without water supply in the area.

Model	Accuracy	True Positive Rate	True Negative Rate	Cut-off point
GWR	71.0%	82.6%	60.4%	0.3
Logistics Regression	76.8%	81.46%	71.6%	0.4

Conclusion and Insights

Determining the exact status of the water pumps by 3 status levels is more challenging compared to binary level. However, by regrouping the status from 3 to 2 levels, important information is lost; E.g. Ministry of Water would not be able to prioritize which water pump to send the maintenance personnel between the "functional needs repair" and the fully "non-functional" water pumps. The recommended model is the Decision Tree model from the resampled dataset.

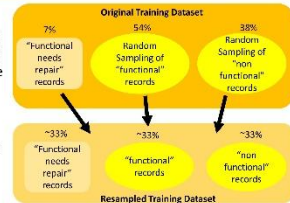
Spatial data does not seem to have a reinforcing effect on the predictive model. This may be due to the limited dataset used for GWR for this project, as there is processing limitation in R. Future work will include more experiments to try out adaptive bandwidth to ascertain if spatial data is useful in this predictive model.

Methodology of Multinomial Logistic Regression and Decision Tree

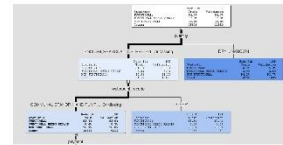
As the status of the water pump took 3 different classes, multinomial logistic regression and decision tree are used to fit the model. One of the response's class is only 7% of the dataset, hence resampling is also performed to reduce noise in the dataset and prior probabilities set to the resampled dataset.

Multinomial Logistic Regression: Generalised logistic regression model for multiple classes. Setting class $j = 0$ as reference, the logit for each $j = 1, \dots, C - 1$ is given as:

$$\log\left(\frac{p(j)}{p(0)}\right) = \alpha(j) + \beta_1^{(j)}X_{1i} + \dots + \beta_{K-1}^{(j)}X_{Ki}$$



Decision Tree is also used to predict the status of the pumps. A maximum depth of 10, and a minimum number of records in each leaf of 5 is set.



Practice Research Paper

- Practice research is a form of academic research which incorporates an element of practice in the methodology or research output.
- Content of a practice research paper
 - Abstract: summarises the content of the research paper, not more than 350 words.
 - Introduction: context-centric and analytical-centric
 - Literature review
 - Analysis process: use appropriate visual to show the process, provide useful tips and tricks on data preparation.
 - Analytical methods: “layman friendly” explanation but mathematically rigorous.
 - Results and Discussion: go beyond accuracy and precision, related the analysis results to real world usage.

Introduction

- **The introduction** explains what you want to do, what are the research gap(s) your research aims to bridge and what are the intended outcomes (if possible). You should include the following:
 - A high-level overview of the topic
 - Explain the background and context of this project
 - Outline the content of the research paper.

Problem Statement and Objectives

- Writing a **problem statement** can help you contextualize and understand the significance of your practice research problem. A problem statement is the central assertion or position that your project will argue for. It can be several paragraphs long and serve as the basis for your practice research project, or it can be condensed into just a few sentences in the introduction section.
- **Objectives** provide a list of goals that will be achieved through the data analytics. You should:
 - describe the precise issue that the project will address
 - show the relevance of the problem
 - set the objectives of the project

Literature Review

- A **literature review** is a survey of scholarly sources on a specific topic. It provides an overview of current knowledge, allowing you to identify relevant theories, methods, and gaps in the existing research. A strong literature review convinces the readers that your project has a solid foundation in existing knowledge or theory. It also shows that you are not simply repeating what other people have already done. You should:
 - compare what are the main theories, methods, debates and controversies
 - Describe what are the strengths and weaknesses of different approaches
 - show how you intend to build on, challenge, or synthesize the work of others

Data and Methods

- This section should describe the overall approach and practical steps you will take to answer your problem statement.
- It defines the research methods and logic steps, such as:
 - which research methods will be used
 - describing the data including the data source

Analysis and Results

- The analysis section is concerned with analyzing the results you have achieved through the data analytics project.
 - Results are actual statements of observations, including statistics, tables and graphs.
 - You should present both negative and positive results with sufficient details so that readers can draw their own inferences and construct their own explanations.

Discussion

- The first paragraph of the discussion section usually summarizes the project findings briefly, and set the stage for the rest of the discussion.
 - You should place your results in context of existing literature on the topic, and describe why they are important.
 - While the results section needs to be objective and factual, the discussion is your chance to interpret your findings. You can discuss why the results are important, how it can be used, etc.

Conclusions and Contributions

- You can explore the potential contributions of the data analytics project for theory or practice, and emphasize what you aim to contribute to existing knowledge on the topic. For example, your project might have contributions for:
 - improving processes in a specific location or field
 - informing policy objectives
 - strengthening a theory or model

References

- All factual material that is not original must be accompanied by a reference to its source.
- Please use the [APA citation style](#). There are also citation management tools such as [EndNote](#) or [Zotero](#), etc.
- Please contact SMU Libraries (library@smu.edu.sg) if you need assistance with citation management tools.

Example of a practice research paper

- https://ink.library.smu.edu.sg/sis_research/2448/

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Time-Series Data Mining in Transportation: A Case Study on Singapore Public Train Commuter Travel Patterns

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Data Analytics Project Poster Day



Data Analytics Project Poster Day



Final Deliverable

- Data Analytics Project poster
 - Must be in ***jpg*** format
 - Name the file **ISSS602_AY2025-26T2_G0-Group00**
- Practice research paper
 - Must be conformed to SAS Global Forum.
 - Not more than 3500 words.
 - in ***MS Word*** format.
 - Name the file **ISSS602_AY2025-26T2_G0-Group00-v1.0.**
- Analytical sandbox: raw data, analytical data and project files.
 - In KNIME workspace
 - Keep all deliverables in a single folder.
 - Name the folder **DAP_G0-Group00**